

① Interview

Qualcomm - RF role (Software)

23-08-2024
S. Mahabub Peer

1. Intro
2. My ^{B.Tech} project is based on textile MIMO antenna, highly isolated.
 - What is Isolation, Why Textile
 - What is Gain, Efficiency.
3. From here Communication subject he asked.
 - Write block diagram of Tx/Rx for passband signal. (coordinates)
 - While I was drawing, each block is asked, What is Sampler, Quantiser, Encoder, DAC, Modulator.
 - Why carrier signal is used in modulation.
4. What is MIMO? Difference between Diversity and MIMO.
5. Some concept in Digital Communication, Beyond ASK, FSK,
6. C Programming - He wants advanced C $\rightarrow$ DS, Graphs, Tree

Q1. unsigned char val = 150;

```
for(unsigned char i=0; i<2*val; ++i){  
    // do something
```

}

How many iterations in for loop? (Not 300).

Q2. In python write for loop.

Q3. Null Pointer vs. Void Pointer

Q4. Why Pointer? , What is purpose of pointer.

Q5. Difference between static and dynamic memory allocation.

Finally Not selected.

② MICRON. (DEG Verification)

28-08-2024
S. Mahaboob Peer.

- ① Introduction.
- ② Draw latch using MUX.
- ③ Derive D-Flip flop using above.
- ④ Explain working of above
- ⑤ What is setup and hold time at transistor level.
- ⑥ From Verilog code. (Based on Event Scheduling)

<pre>always @(*) begin a <= 0; b <= 0; c <= 0; \$display(a,b,c); end</pre>	<pre>always @(*) ^{begin} \$display(a,b,c) → ? a <= 0 ; \$display(a,b,c); → ? #5 b <= 1 ; \$display(a,b,c); → ? #10 c <= 2; \$display(a,b,c); → ? end</pre>
---	--

- ⑦ Inter^{delay} and Intra^{blocking} for blocking & non-blocking statements. (Read blocking & non-blocking from pdf given in resources)

```
always @(*) begin
    a <= 0;
    a = 5;
    $display(a); $strobe(a); end
```

Finally NOT SELECTED.

(~1 hour).

③ Mediatek. (Protocol Stack Engineer)

05-09-2024

S. Mahaboob Peer

① Introduction

② They started asking questions for Communication Systems.

③ Draw block diagram of bandpass communication

They started questioning on each block in detail.

④ Analog & Digital modulation techniques. Explain them.

⑤ What are different types of noises. ~~W~~

Is there any useful noise? (I said PN ~~noise~~ sequence)

⑥ Line coding, Source Coding, Channel Coding.

⑦ Project (Design of highly isolated MIMO antenna for 5G application)
(substrate) which material is used. What is gain, isolation.

Define diversity, efficiency, gain and directivity.

Difference b/w gain and efficiency.

⑧ Basics of BWT.

Evolution of 1G, 2G, 3G, 4G & 5G.

⑨ 3GPP, GSM and related.

(40 min)

⑩ OT based ~~an~~ ci) C-Programming (3 problem statements given have to write code)

(ii) Signals & Systems, COA

GATE level questions.

(iii) Aptitude, Digital

(iv) C-Program Snippet

④ Synopsys (Digital Role).

11-09-2024
S. Mahaboob Peer

- ① Introduction.
- ② Write Draw Waveform of CMOS inverter.
- ③ In which region it will be worked as switch & Amplifier.
- ④ They shortlisted for Digital role but their requirement is Analog and Mixed role.
- ⑤ They mostly asked questions on Analog Electronics only.

Finally NOT SELECTED.

⑤ Analog Devices (^{1st} Online interview). (Digital Role)

14-11-2024
S. Mahabub Peer.

2 rounds to all selected candidates after OT.

① Introduction

② Projects (^{1st} - Asynchronous FIFO ^{2nd} - 8x8 Vedic Multiplier)

They asked to explain both verilog projects (not verilog code) in detail.

③ From STA.

- (i) Why STA and what is it?
- (ii) Setup time, hold time, skew and all.
- (iii) If clock delay & Datapath delay how it will be affected on setup & hold for a system.

④ C-Programming

- (i) Take an array of size n and make last 3 elements 0.
- (ii) How to remove last 3 elements?
- (iii) Write program to sum of n integer numbers.

⑤ Aptitude & Puzzle.

- (i) Formula for sum of n numbers. Even no., Odd no.
- (ii) Measure 4L using only 3L & 5L jars.

(~ 20min)

2nd round

① Introduction

② Write verilog code for ' $\frac{f}{4}$ ' frequency divider.

hint: using Counters

③ Questions on calendar (Aptitude)

④ Write 'C' program for calendar if Day 1 is started with specific day then what day will be after 'N' days (N=10).

⑤ Simple Digital Electronics Questions (Combinational).



Finally NOT SELECTED.

⑥ C-DOT (1st interview)

(Online) (Communication + Digital Role)

Panel - 3 members (~30 min)

28-11-2024

S. Mahabooob Peer

① Introduction.

② Projects (2 verilog, 2 Antenna) (15 min)

1st Project Asynchronous FIFO. 16x8

2nd project 8x8 Vedic multiplier (MAC)

} verilog

Area of interest
I kept in Resume

HSCN

Digital Electronics

STA

BVLSI

(i). Why Vedic multiplier. How it is different from conventional multiplier.

I explained procedure and I used Ripple Carry Adder (RCA)

(ii). Why RCA? What are others? (I said Carry Look Ahead)

What's difference between them.

③ Why FIFO? What are applications?

(No Verilog code asked).

④ From Here Analog Electronics asked.

(i) What is capacitor?

(ii) If voltage source directly given to "C" what will happen?

⑤ Antenna basic. (i) What is directivity, Efficiency, gain?

⑥ From Verilog-Vivado.

(i). What process will be done in Vivado (Simulation, Synthesis)

(ii) Difference b/w Compiler & Interpreter.

⑦ From here Signals & Systems and Control Systems (Basics)

(i) What is Transfer function? & its significance.

(ii) What are poles? How stability will be defined?

(iii) What is Fourier Series, Fourier Transform and D.F.T

(iv). F.T of Sin wave.

8) Again Analog Electronics.

(i) How poles will say oscillatory?

(ii) Design an oscillator using OP-Amp.

(iii) What is CMRR? What is purpose of CMRR?

9) What is Sampling? What is Aliasing?

10) What are Analog modulation & Digital Modulation techniques?
What will be carrier used in both?

11) Super heterodyne Receiver & block diagram.

12) In 5G which technique will be used

(Extended OFDMA)

(i) What are S-Parameters

(Project Antenna related)

(ii) MIMO communication receiver.

C-DOT (2nd Interview) (Offline)

Panel consists of 3 members (~30min).

9-12-2024

S. Mahaboob Peer

- ① Introduction.
- ② Projects Related Questions (I did 2 Verilog & 1 Antenna Design in B.Tech and another Antenna design in M.Tech).
Total = 4 projects.
- ③ They mainly asked my antenna projects. My antenna project is ^(B.Tech) Design of highly isolated MIMO antenna for 5G application.
 - i. What are the parameters are required to design an antenna.
I said frequency, Bandwidth, but they expected to tell wavelength
 $(\lambda = \frac{c}{f})$: for dipole antenna $(\frac{\lambda}{2})$. (Antenna)
(AWP)
 - ii. Why MIMO, I said to improve diversity Gain.
 - iii. What is Diversity, Path loss, fading, multi-path? (BWT)
- ④ Next, Wireless Communication. What is MIMO communication
What is channel and Noise. $\Rightarrow$ Types of Noises. & channels.
- ⑤ When ^{different} 3 types of frequencies are there in a signal. How will you detect them?
(Ans: Fourier Transform)
- ⑥ From here another panel member asked Digital Communication.
 - i. What are Digital Modulation techniques. I said ASK, BPSK, FSK, QPSK, QAM.
 - ii. Draw waveform and explain ASK, BPSK and FSK.
 - iii. Why BPSK, QPSK & QAM. Why they went from BPSK to QAM.
 - iv. What are symbol and bits in above.

I explained them with constellation plots

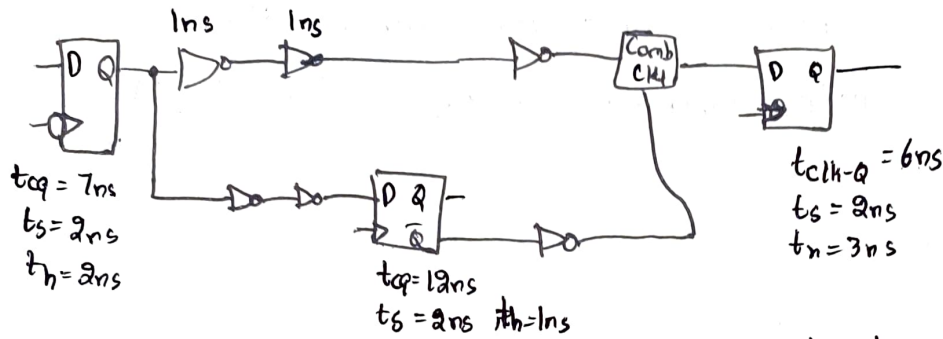
16-01-2025

7 Ansys Interview Experience (1hr)

Shaik Mahaboob Peer
208123027 -CS

1 Brief introduction

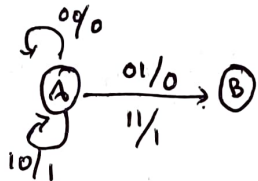
2 Calculate Max. clk Freq.



In this question he discussed all STA related topics, Setup, Hold & their violations, half path delay and so on.

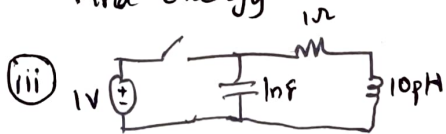
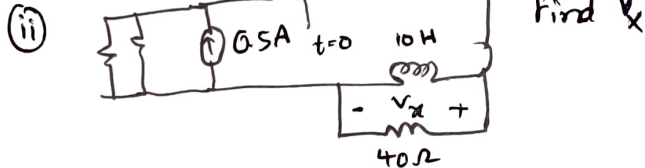
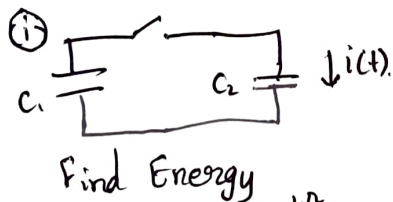
3 Design FSM to when burst input bit streams then divide by 2.

Ans. like



like this way: take 2 bits & do FSM

4 From OT few Analog questions he asked.



Find current through inductor
at $t=0^+$ & $t=\infty$ (steady state)

5 As I kept project on Antenna (Ansys HFSS)

(i) What is 50Ω impedance

(ii) What is impedance for coaxial cable (70Ω)

(iii) How many conductive layers in coaxial cable. (Basic EMIC & EMTL questions)

6 No Questions asked from Project except above that too on Ansys related.

⑦ Then he jumped into BVLSI.

① Draw nMOS

② Explain in detail of nMOS.

③ Why Saturation current is not zero when there is no depletion region after pinch off voltage. (Think in PN junction diode concept?)

⑧ ASIC Design Flow.

That's it.

Next Normal HR round (No elimination for us only)

Finally Placed 🎉 😊



(SMP)